



❖ *A Clinical Decision Support Tool for Breast Cancer Detection*

Reducing Missed Diagnoses with AI

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□ The Clinical Problem:

- ❖ *False negatives in cancer screening lead to delayed treatment and worse patient outcomes.*
- ❖ *Manual screening is time-consuming and subject to human fatigue/error.*
- ❖ *Late-stage treatment costs the healthcare system significantly more than early intervention.*



□ *Our AI Solution:*

- ❖ *An AI-powered "second pair of eyes" for radiologists.*
- ❖ *Analyzes 30 tumor characteristics instantly.*
- ❖ *Flags high-risk cases for priority review.*



□ *Performance at a Glance:*

- ❖ *98% Recall: Correctly identifies 98% of malignant tumors.*
- ❖ *99.5% ROC-AUC: Exceptional ability to distinguish between benign and malignant.*
- ❖ *"Our model misses fewer cancers than the current standard of care."*



❑ *Business Impact (ROI):*

- ❖ *Estimated \$100k savings per 1000 patients by catching cancer earlier.*
- ❖ *Reduced unnecessary biopsies (High Precision).*
- ❖ *Faster triage times, improving hospital throughput.*



□ Ethical AI & Fairness:

- ❖ *We conducted a rigorous bias audit.*
- ❖ *Disparate Impact Ratio: 1.09 (Passes the 80% rule).*
- ❖ *The model performs equally well for small and large tumors, ensuring equitable care for all patients.*



❑ Implementation Roadmap:

- ❖ Phase 1 (Pilot): Deploy in a single radiology department.
- ❖ Phase 2 (Integration): Connect API to the Electronic Medical Record (EMR).
- ❖ Phase 3 (Scale): Hospital-wide rollout with continuous monitoring.



❑ Risk Management:

- ❖ Risk: False Positives (patient anxiety).
- ❖ Mitigation: High precision means low false positives. Human-in-the-loop (HITL) sign-off required.
- ❖ Risk: Algorithmic Bias.
- ❖ Mitigation: Quarterly bias audits and continuous data monitoring.



Thank you for your time.

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